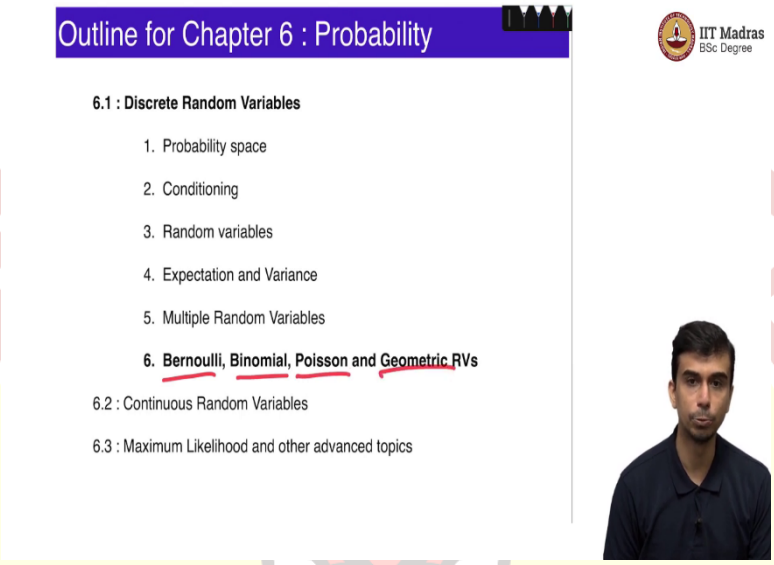



# IIT Madras

## ONLINE DEGREE

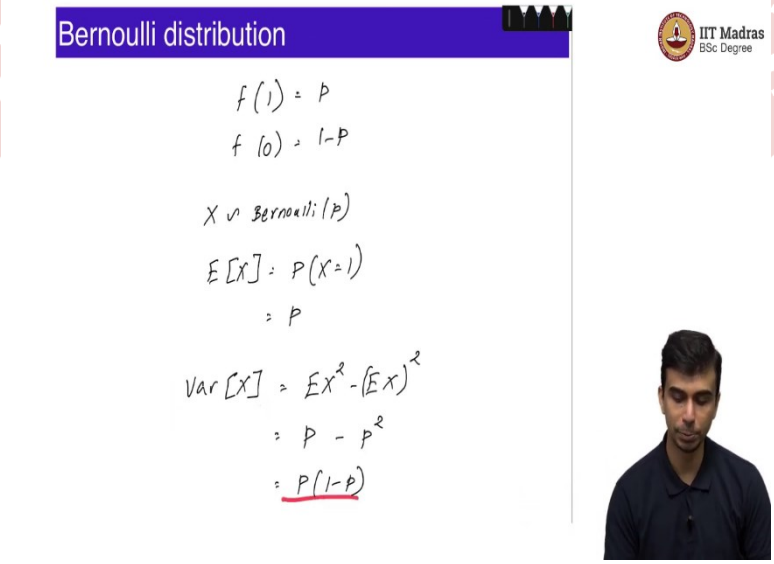
**Machine Learning Foundations**
**Assistant Professor Harish Guruprasad Ramaswamy**
**Department of Computer Sciences and Engineering**
**Indian Institute of Technology, Madras**
**Bernoulli, Binomial, Poisson and Geometric Random Variables**

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Now we will wrap up this week of discrete random variables by giving a list of standard random variables that you would see in text, in books all over, which are the Bernoulli, the Binomial, the Poisson and the Geometric random variables. All of these are discrete random variables. We will give their probability mass function and some of their moments. We will give their expectation, and if possible, the variance also.

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So what is the Bernoulli distribution? Bernoulli's distribution is the simplest distribution that you can think of, which is, that will give the probability mass function of that, which is f of 1 = p, and f of 0, = 1 - p. So this is a probability mass function which can take only two values, 1 and 0, and it is parameterized by p.

For example, if you had a coin whose probability of heads was p, small p, then that would correspond to a Bernoulli random variable, or Bernoulli distribution with bias p. So this p is often called the bias. So if p = half, it is an unbiased Bernoulli distribution, but you can have a general p, which would be called the bias of the Bernoulli distribution. So this is a pretty simple distribution, we can take only values 1 and 0.

What is the expected value of, let us say X, you might see written as, X is distributed as Bernoulli p. This might be a statement that you would see. X is a random variable is distributed as Bernoulli p. So by that we mean the probability mass function of X = this.

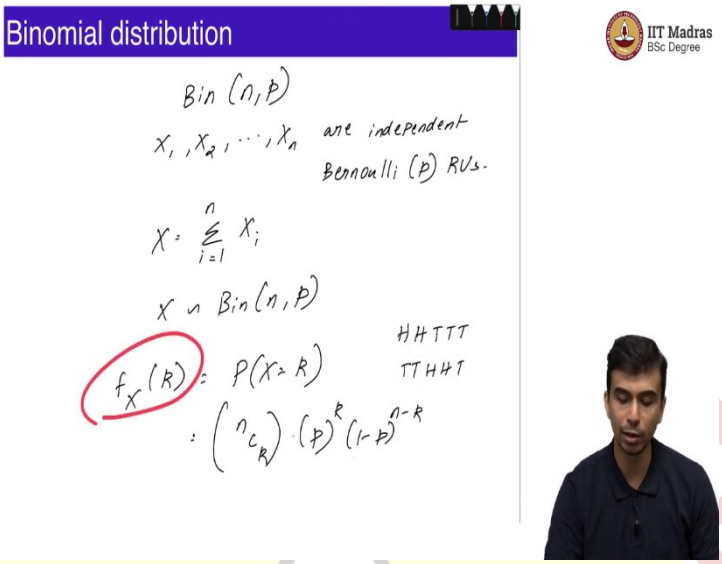
So what is the expected value, if X is distributed as Bernoulli p, what is the expected value of X? Well, that = probability of x = 1, which is exactly = p. So if X is distributed as Bernoulli p, the expected value of X is p.

What is the variance of X? We know variance of X, expected value of X squared - expected value of X, the whole square. But X is a random variable which takes values only 1 or 0. So X and X squared are really the same. So E X squared = E X, so which is p, p - p squared, which = p × 1 - p. So this is instructive to look at.

The average or the mean of X is simply p, but the variance of X is p × 1 - p. So this is, recall we had said variance as the measure of deviation of the random variable from its mean. So if p was either very small or very large, which correspond to p being very close to 1 or very close to 0, the random variable is always going to take either always 1 or always 0, so which means that your variance is going to be low.

So if you want to, the value of p which maximizes the variance is the value of p which maximizes p × 1 - p, and that is exactly = half. When, p = half is when the variance of the Bernoulli random variable is maximized. That is all about for a Bernoulli distribution.

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Now we have the binomial distribution. The binomial distribution is denoted as binomial of n comma p. And the binomial distribution occurs naturally when you add independent Bernoulli random variables. Let us say you have x 1 x 2 dot, dot, dot x n are independent Bernoulli p random variables. And you can define X to be = sum over i = 1 to n, X i. Each of these are Bernoulli random variables and they are independent and you add all of them to get X.

In this case X would be distributed as binomial of n comma p. That is fine, but what is the probability mass function of X? What is the probability mass function of X? Well f X of k is probability of X = k, which is when will, let us, to interpret it this way, if you are tossing n coins with bias p, what is the probability that the number of heads = k?

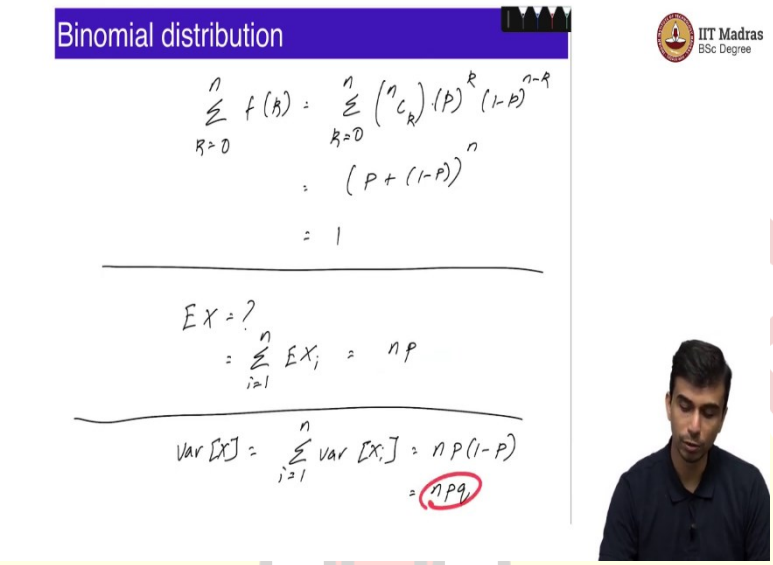
Well there are 2 power n possibilities, because there are 2 power n possible coin toss sequences out of which, which of them give you exactly k coin tosses, just n, there are n choose k ways in which you can get exactly k heads. And each of those correspond to a sequence where something like let us, say k = 2 would mean H H T T T T T. H H can be anywhere. It can be for example T T H H T or so on.

But the key thing to note is any, the probability of any of this exact sequence happening is that there are k heads and n - k tails. The (prob), and these are independent. The probability that you have exactly that sequence is exactly = p power k × 1 - p power n - k.

So this is why you can see that it is called the binomial distribution because it contains the binomial coefficients. The n choose k comes from the fact that you can, if you want, you have n coin tosses and you want k heads there are, you have to choose the k's parts out of the n's parts.

So there are n choose k ways in which you can do that and the probability of each of those coin toss sequences is exactly = p power k × 1 - p power n - k. So that is your probability mass function of the binomial random variable.

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What is the expected value of, let us say you have X is distributed as, well, first of all, let us check whether it is a valid probability mass function? So which means that what is, is sum over k = 0 to n. So the maximum value of the binomial random variable can take is n, because even if all the n tosses turn out to be heads, that is the maximum value it can take. So k equals 0 to n f of k.

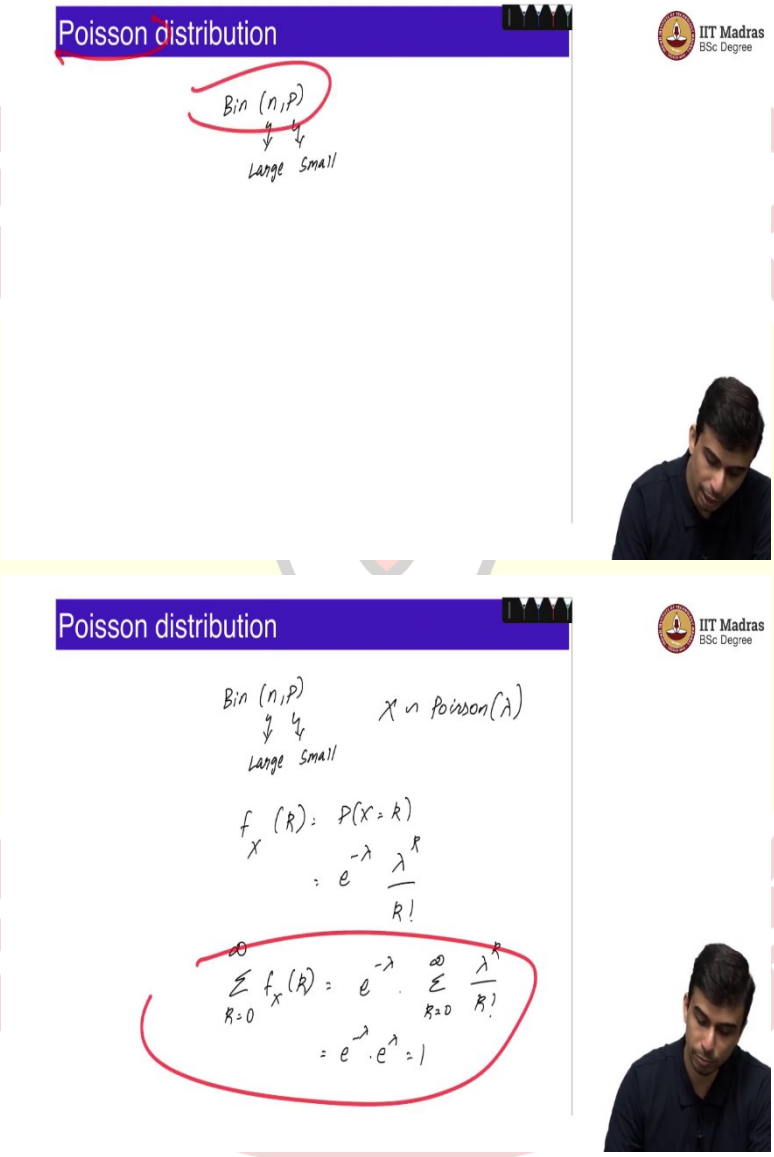
What is that? Well that = sum over k = 0 to n, n choose k × p power k × 1 - p power n - k. If you know your binomial theorem, this is exactly = p + 1 - p power n, which is exactly = 1. So you have that this is a valid probability mass function. So that is the first, that is the first property.

Then, then we move on to expectation. What is the expected value of, what is expected, let us say X is distributed as binomial of n comma p, what is E X? Here, you can use the linearity of expectation, that is, E X is simply = some over i = 1 to n, E X i. Each of these are Bernoulli random variables, so which, Bernoulli random variables expectation = p, so this = n p.

What about variance of X? This requires an important property of variance, that is, variance of, if, variance of the sum = the sum of the variance, if the terms are independent. If you have variance of the sum of independent random variables = the sum of the variance.

In this case they do happen to be independent, so variance of x = sum over i = 1 to n variance of X i. And variance of X i is p ×1 - p. So you have n such terms. Put all of that together you get n p 1 - p. Sometimes you might call, you might see 1 - p called as q, which is n p q. That is another standard way. So that means that we have the mean of the binomial distribution is n p, and the variance is n p q.

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So we finally move on to the Poisson distribution. The Poisson distribution is a specialized, it can be viewed as a specialized case of binomial distribution. It is the probability of, it is effectively the random variable of very rare, corresponding to the very rare events.

For example, let us say you have a factory manufacturing products. Let us say the failure rate of is, let us say 10 power - 5. One in ten thousand products fails. And let us say you manufacture

mil, or millions or even tens of millions of products every year. You want to know how many failures happen in a year.

This is an example. You can model this using binomial random variables or binomial distribution, but it is easier to use Poisson because of the very small failure probabilities, and, very small probabilities p, and large number of n.

So for example, this is the, if binomial of n comma p, if n is very large and p is very small, this corresponds to a special case which approximates the Poisson distribution. But we can define the Poisson distribution independently, that is f x of k is probability of x = k, which is e power - lambda to lambda power k by k factorial. X is distributed as Poisson, and the Poisson distribution parameter is lambda.

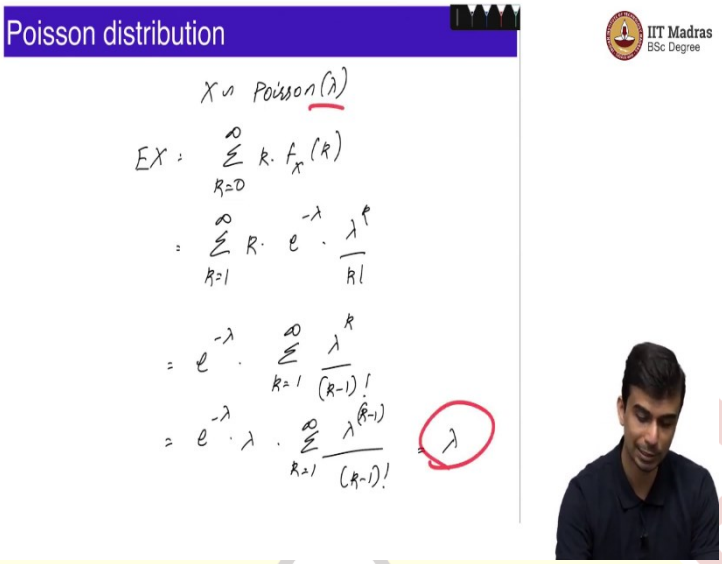
The same way for the Bernoulli distribution, the parameter is p, and for the binomial distribution its parameters by n and p, for a Poisson there is a parameter lambda, which is a positive value. It can be, it is a real valued number.

The probability of, if X is distributed as Poisson of lambda, then the probability mass function f X of k is given by probability of X = e power - lambda, lambda power k by k factorial. We can immediately see that this does sum to 1 as you sum over infinity, that sum over k = 0 to infinity f X of k.

So this is the first discrete random variable that you are seeing, which can take, it is discrete, still, it can take only integer numbers, but it can take arbitrarily high values. So you can go all the way to infinity.

So sum over k = 0 to infinity f x of k is e power - lambda × sum over k = 0 to infinity lambda power k by k factorial. If you know Taylor series, this is exactly = e power lambda. This is e power - lambda × e power lambda = 1. So this does sum to 1. So that property is satisfied.

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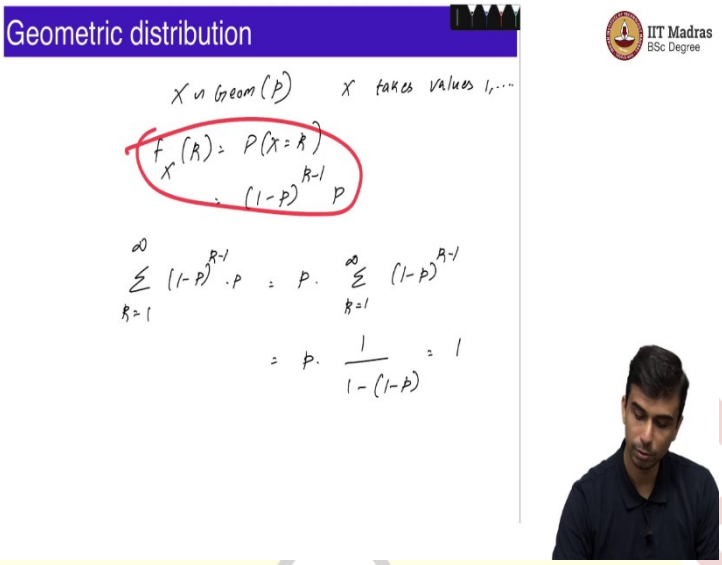

What about the expectation and variance? We will just do the expectation of the Poisson distribution. It is pretty easy to do. Let us say X is distributed as Poisson of lambda. What is E X? Well, E X is sum over k = 0 to infinity, k times f x of k, which is sum over k = 0 to infinity. We will just make it 1 to infinity because k = 0 adds nothing.

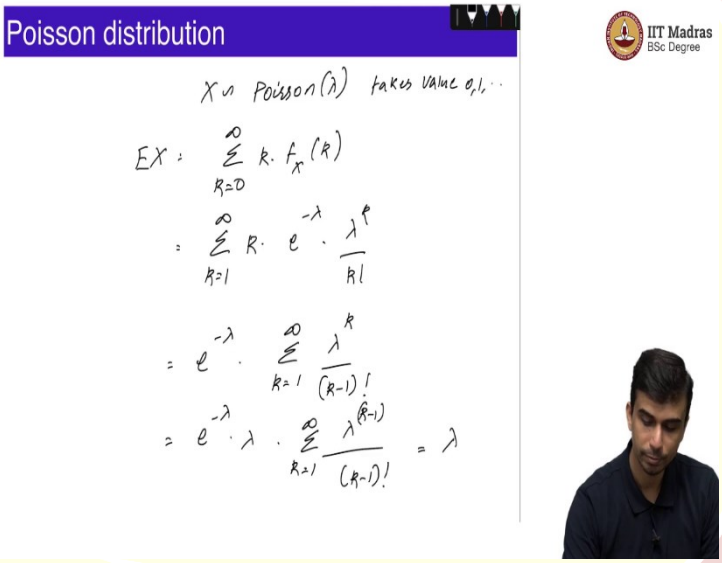
So k = 1 to infinity k times e power - lambda, lambda power k by k factorial. And k factorial and k cancels to get k - 1 factorial. So we will take e power - lambda outside. e power - lambda × sum over k = 1 to infinity lambda power k by k - 1 factorial.

Let us take just 1 lambda outside. So e power - lambda × lambda × sum over k = 1 to infinity lambda power k - 1 by k - 1 factorial. Let us call k - 1 a dummy variable by let us say l, which means that k = 1 to infinity is the same as l = 0 to infinity lambda power l by l factorial which is exactly = e power lambda. So e power lambda cancels with e power - lambda to give us lambda.

So the expected value of a Poisson distributed random variable with parameter lambda = lambda. In fact, the variance is also = lambda. However, you have to multiply, k, you have to have k squared, here you have to get e square, you have to get k squared, and then you have to do this two times, but in, technically it is the same thing. The variance E X squared = lambda squared + lambda, so E X squared - E X the whole squared becomes lambda. So the variance of a Poisson is also = lambda.

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And finally, we will move to the last of our distributions, which is the geometric distribution. The geometric distribution is, how to motivate that? Well, let us say you are tossing a coin with bias p. This is the experiment. You keep on tossing the coin until you get a head.

Let us say if you, at the first time itself you get a head, you stop there, or you keep on tossing the coin till you get a head. The number of coin tosses you make is geometrically distributed. So that means that if X is distributed as geometric with bias p, so the parameter, the bias of the coin exact is the same as the parameter of the geometric.

We will say X is distributed as geometric with p if f X of k, which is probability of x = k, what is the probability that you need exactly k coin tosses to see a head for the first time. So which means that the first k - 1 times, you should have seen a tail. The probability of that happening is 1 - p power k - 1.

Why is that? Because these are, k - 1 times you should independently see a tail. So it is 1 - p power k - 1. And at the kth position, and the kth toss, you should see a, a head. The probability of that happening is p. So f X of k is 1 - p power k - 1 × p.

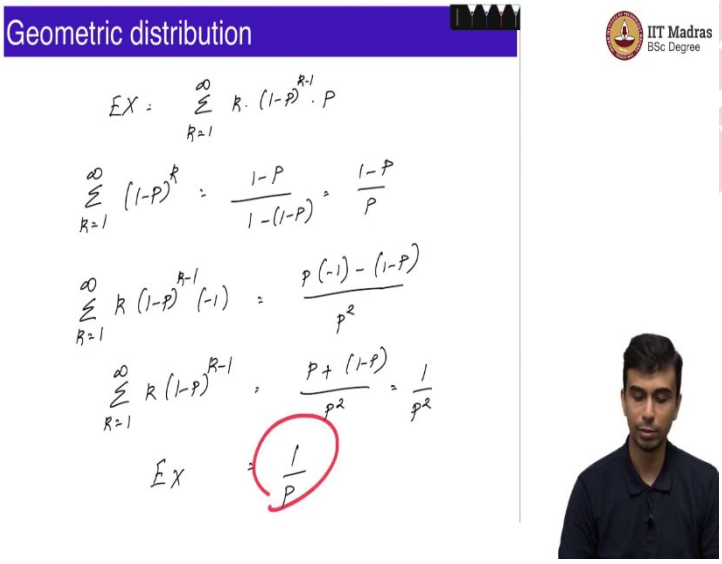
So this is the probability mass function of the geometric random variable. And X takes values in 1 to infinity. Poisson takes values 0 to infinity, integers only, 0, 1, dot, do, dot. Geometric, however, takes values, cannot take a value of 0. I mean the number of coin tosses can never be 0 to get a single head. The best case scenario, you can, you will have to still toss a coin one time. So X takes values 1, 2, 3, all the way to infinity.

First of all, once again, you want to check whether there is a valid probability mass function or not. We will do that. Sum over k = 0, sorry, 1 to infinity 1 - p power k - 1 × p = p × sum over k = 1 to infinity, sorry, 1 to infinity 1 - p power k - 1.

So this is a geometric series and by standard geometric series formula this = p × 1 by 1 - 1 - p, this = 1. So you have that this does sum to 1. So this is a valid probability mass function. So that is the first thing out of the way. Now let us try to see if we can get the expectation of this random variable.

Intuitively what should you expect? So p is the bias of the coin. So if p is very low, if, let us say you have a super rare, you have a coin where heads is super rare, then you expect to toss the coin more times. So you expect the, expected, I mean, you, you would, the expectation of the number of coin tosses should scale inversely with the bias.

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Let us see if that is really true. The, our intuition says that if p is very small then you should toss more coins. So the expected value of this random variable should be inversely dependent on p. Let us try to get that.

So what is E X? Well, E X is sum over k = 1 to infinity k × 1 - p power k - 1, × p. So this is not so easy to do straight forward because this is not a geometric series, it is not arithmetic series also, this is what is called an arithmetico-geometric series. You can use some standard tricks to get this sum. The standard trick to do that is to use differentiation of a series.

Let us say you have sum over k = 1 to infinity 1 - p power k. What is that? Well, you know by geometric series, this is exactly = 1 - p by 1, by 1 - 1 - p, which = 1 - p by p. This is a fact of geometric series. Let us differentiate this expression, left and right side by p, with respect to p.

If you do that, sum over k = 1 to infinity, k × 1 - p power k - 1 × - 1 = p squared, p × - 1 - 1 - p. This is u by v, this v squared v d u - u d v.

The reason, you can see, is obvious that you are getting sum over k = 1 infinity k × 1 - p power k - 1, which is the term that you want for E X also. So let us just multiply by - 1 on both sides to get what we want, sum more k = 1 to infinity k × 1 - p power k - 1, is simply = p -, sorry, p + 1 - p by p square, which = 1 by p squared.

So sum over k = 1 to infinity k × 1 - p power k - 1 is 1 by p squared. But what do we want? We want sum over k = 1 infinity k × 1 - p power k - 1 multiplied by p. So we have that E X is simply = 1 by p, which does agree with our intuition, that is, if p is small, you expect more number of coin tosses to get to the head.

You can define, you can get the variance also in the same way. You get E X squared, but you might need to differentiate twice to get this thing. But it is a very simple logic. And you can get the variance also in the same way. But the variance is not very important here. The more important thing is that the mean of a geometric distribution is 1 by p.

So with that you have the basic tools required for handling discrete random variables. In the next week we will move to continuous random variables. But that is all for discrete random variables.